

AN AI POWERED SOFTWARE-BASED FIRE DETECTION SYSTEM FOR CCTV PLATFORMS.

REST API DOCUMENTATION

Version: v1 **Base URL:** <https://firemex.netlify.app/api/v1>

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Overview

This API provides programmatic access to a AI - fire detection platform, covering authentication, user management, camera devices configuration, alert management, incident management, report generation, AI model configuration, and notifications.

Conventions

- All request/response bodies are **application/json** unless otherwise noted.
- All timestamps are ISO 8601 UTC (e.g. **2026-07-29T10:15:00Z**).
- All list endpoints support pagination via **page** and **limit** query parameters.

All authenticated endpoints require a Bearer token in the **Authorization** header:

Authorization: Bearer [access_token]

1. User Authentication

1.1 Login

POST /api/v1/auth/login

Authenticates a user and returns an access/refresh token pair.

Request Body

Field	Type	Required	Description
email	string	Yes	User's email address
password	string	Yes	User's password

```
{
  "email": "user@example.com",
  "password": "SecurePassword123!"
}
```

Response **200 OK**

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIs..."
}
```

```
"refresh_token": "d1b2c3a4-....",
"token_type": "Bearer",
"expires_in": 3600,
"user": {
  "id": "usr_001",
  "email": "user@example.com",
  "role": "admin"
}
}
```

Errors: 400 Bad Request
401 Unauthorized

1.2 Refresh Token

POST /api/v1/auth/refresh

Exchanges a valid refresh token for a new access token.

Request Body

Field	Type	Required	Description
refresh_token	string	Yes	Previously issued refresh token

```
{
  "refresh_token": "d1b2c3a4-...."
}
```

Response 200 OK

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIs...\"",
  "expires_in": 3600
}
```

Errors: 401 Unauthorized
403 Forbidden

1.3 Logout

POST /api/v1/auth/logout

Invalidates the current access/refresh token pair.

Headers: Authorization: Bearer [access_token]

Response: 204 No Content

Errors: 401 Unauthorized

2. Users

2.1 List Users

GET /api/v1/users

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
role	string	-	Filter by role (admin, operator, viewer)
search	string	-	Search by name or email

Response 200 OK

```
{
  "data": [
    {
      "id": "usr_001",
      "name": "Jane Doe",
      "email": "jane@example.com",
      "role": "admin",
      "status": "active",
    }
  ]
}
```

```
    "created_at": "2026-01-10T08:00:00Z"
  },
],
"pagination": {
  "page": 1,
  "limit": 20,
  "total": 42,
  "total_pages": 3
}
}
```

2.2 Create User

POST /api/v1/users

Request Body

Field	Type	Required	Description
name	string	Yes	Full name
email	string	Yes	Unique email address
password	string	Yes	Initial password
role	string	Yes	admin, operator, or viewer

```
{
  "name": "John Smith",
  "email": "john@example.com",
  "password": "TempPass123!",
  "role": "operator"
}
```

Response 201 Created

```
{
  "id": "usr_045",
  "name": "John Smith",
  "email": "john@example.com",
}
```

```
"role": "operator",
"status": "active",
"created_at": "2026-07-29T10:00:00Z"
}
```

Errors: 400 Bad Request, 409 Conflict (email already exists)

2.3 Get User

GET /api/v1/users/{id}

Path Parameters: id (string, required) - User ID

Response 200 OK

```
{
  "id": "usr_001",
  "name": "Jane Doe",
  "email": "jane@example.com",
  "role": "admin",
  "status": "active",
  "created_at": "2026-01-10T08:00:00Z",
  "last_login": "2026-07-28T14:22:00Z"
}
```

Errors: 404 Not Found

2.4 Update User

PUT /api/v1/users/{id}

Path Parameters: id (string, required)

Request Body

Field	Type	Required	Description
name	string	No	Full name
email	string	No	Email address

role	string	No	Role
status	string	No	active or disabled

```
{
  "name": "Jane A. Doe",
  "role": "operator"
}
```

Response 200 OK - returns the updated user object.

Errors: 400 Bad Request, 404 Not Found

2.5 Delete User

DELETE /api/v1/users/{id}

Path Parameters: id (string, required)

Response 204 No Content

Errors: 404 Not Found, 409 Conflict (cannot delete last admin)

3. Cameras

3.1 List Cameras

GET /api/v1/cameras

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
status	string	-	online, offline, maintenance
location	string	-	Filter by site/location name

Response 200 OK

```
{
  "data": [
    {
      "id": "cam_012",
      "name": "Front Entrance",
      "location": "Building A - Lobby",
      "status": "online",
      "stream_url": "rtsp://stream.yourdomain.com/cam_012",
      "resolution": "1920x1080",
      "created_at": "2026-02-01T09:00:00Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 15,
    "total_pages": 1
  }
}
```

3.2 Register Camera

POST /api/v1/cameras

Request Body

Field	Type	Required	Description
name	string	Yes	Display name
location	string	Yes	Physical location
stream_url	string	Yes	RTSP/HTTP stream URL
model	string	No	Camera hardware model

```
{
  "name": "Loading Dock",
  "location": "Building B - Rear",
  "stream_url": "rtsp://192.168.1.50:554/stream1",
}
```



```
"model": "HikVision DS-2CD2xx"
}
```

Response 201 Created

```
{
  "id": "cam_013",
  "name": "Loading Dock",
  "location": "Building B - Rear",
  "status": "offline",
  "stream_url": "rtsp://192.168.1.50:554/stream1",
  "created_at": "2026-07-29T10:05:00Z"
}
```

Errors: 400 Bad Request, 409 Conflict (duplicate stream URL)

3.3 Get Camera

GET /api/v1/cameras/{id}

Path Parameters: id (string, required)

Response 200 OK

```
{
  "id": "cam_012",
  "name": "Front Entrance",
  "location": "Building A - Lobby",
  "status": "online",
  "stream_url": "rtsp://stream.yourdomain.com/cam_012",
  "resolution": "1920x1080",
  "assigned_models": ["model_007"],
  "created_at": "2026-02-01T09:00:00Z"
}
```

Errors: 404 Not Found

3.4 Update Camera Details

PATCH /api/v1/cameras/{id}

Path Parameters: id (string, required)

Request Body (partially updated request)

Field	Type	Description
name	string	Display name
location	string	Physical location
status	string	online, offline, maintenance
stream_url	string	Stream endpoint URL

```
{  
  "status": "maintenance"  
}
```

Response 200 OK - returns the updated camera object.

Errors: 400 Bad Request, 404 Not Found

3.5 Delete Camera

DELETE /api/v1/cameras/{id}

Path Parameters: id (string, required)

Response 204 No Content

Errors: 404 Not Found

4. Alerts

4.1 List Alerts

GET /api/v1/alerts

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
severity	string	-	low, medium, high, critical
status	string	-	new, acknowledged, resolved, dismissed
camera_id	string	-	Filter by originating camera
from	string	-	ISO 8601 start date filter
to	string	-	ISO 8601 end date filter

Response 200 OK

```
{
  "data": [
    {
      "id": "alt_1042",
      "camera_id": "cam_012",
      "type": "motion_detected",
      "severity": "medium",
      "status": "new",
      "confidence": 0.92,
      "triggered_at": "2026-07-29T09:58:00Z",
      "thumbnail_url": "https://cdn.yourdomain.com/alerts/alt_1042.jpg"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 8,
    "total_pages": 1
  }
}
```

4.2 Get Alert

GET /api/v1/alerts/{id}

Path Parameters: `id` (string, required)

Response 200 OK

```
{
  "id": "alt_1042",
  "camera_id": "cam_012",
  "type": "motion_detected",
  "severity": "medium",
  "status": "new",
  "confidence": 0.92,
  "triggered_at": "2026-07-29T09:58:00Z",
  "thumbnail_url": "https://cdn.yourdomain.com/alerts/alt_1042.jpg",
  "video_clip_url": "https://cdn.yourdomain.com/alerts/alt_1042.mp4",
  "model_id": "model_007",
  "metadata": {
    "bounding_box": [120, 80, 340, 460]
  }
}
```

Errors: 404 Not Found

4.3 Update Alert

`PATCH /api/v1/alerts/{id}`

Used to acknowledge, resolve, or dismiss an alert.

Path Parameters: `id` (string, required)

Request Body

Field	Type	Required	Description
status	string	Yes	acknowledged, resolved, or dismissed
notes	string	No	Optional reviewer notes

```
{
  "status": "acknowledged",
```

```
"notes": "Reviewed footage, false positive from a shadow."
}
```

Response 200 OK - returns the updated alert object.

Errors: 400 Bad Request
404 Not Found

5. Incidents

5.1 List Incidents

GET /api/v1/incidents

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
status	string	-	open, investigating, closed
priority	string	-	low, medium, high, critical

Response 200 OK

```
{
  "data": [
    {
      "id": "inc_0231",
      "title": "Unauthorized access - Loading",
      "status": "investigating",
      "priority": "high",
      "linked_alerts": ["alt_1042"],
      "assigned_to": "usr_001",
      "created_at": "2026-07-29T10:02:00Z"
    }
  ],
}
```

```
"pagination": {  
  "page": 1,  
  "limit": 20,  
  "total": 5,  
  "total_pages": 1  
}
```

5.2 Create Incident

POST /api/v1/incidents

Request Body

Field	Type	Required	Description
title	string	Yes	Short incident title
description	string	No	Detailed description
priority	string	Yes	low, medium, high, critical
linked_alerts	string[]	No	Alert IDs related to this incident
assigned_to	string	No	User ID of assignee

```
{  
  "title": "Unauthorized access - Loading Dock",  
  "description": "Individual seen entering restricted area after hours.",  
  "priority": "high",  
  "linked_alerts": ["alt_1042"],  
  "assigned_to": "usr_001"  
}
```

Response 201 Created

```
{  
  "id": "inc_0231",  
  "title": "Unauthorized access - Loading Dock",  
  "status": "open",  
  "priority": "high",  
}
```

```
"created_at": "2026-07-29T10:02:00Z"
}
```

Errors: 400 Bad Request

5.3 Get Incident

GET /api/v1/incidents/{id}

Path Parameters: id (string, required)

Response 200 OK

```
{
  "id": "inc_0231",
  "title": "Unauthorized access - Loading Dock",
  "description": "Individual seen entering restricted area after hours.",
  "status": "investigating",
  "priority": "high",
  "linked_alerts": ["alt_1042"],
  "assigned_to": "usr_001",
  "timeline": [
    {
      "event": "created",
      "at": "2026-07-29T10:02:00Z",
      "by": "usr_001"
    }
  ],
  "created_at": "2026-07-29T10:02:00Z",
  "updated_at": "2026-07-29T10:15:00Z"
}
```

Errors: 404 Not Found

5.4 Update Incident

PATCH /api/v1/incidents/{id}

Path Parameters: id (string, required)

Request Body (partial update)

Field	Type	Description
status	string	open, investigating, closed
priority	string	Priority level
assigned_to	string	Reassign to a different user
resolution	string	Resolution notes (required when closing)

```
{  
  "status": "closed",  
  "resolution": "Confirmed authorized contractor; access logs verified."  
}
```

Response 200 OK - returns the updated incident object.

Errors: 400 Bad Request, 404 Not Found

6. Reports

6.1 List Reports

GET /api/v1/reports

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
type	string	-	daily, weekly, monthly, custom

Response 200 OK

```
{
```



```
"data": [
  {
    "id": "rep_089",
    "title": "Weekly Security Summary",
    "type": "weekly",
    "status": "completed",
    "generated_at": "2026-07-28T00:00:00Z",
    "file_url": "https://cdn.yourdomain.com/reports/rep_089.pdf"
  }
],
"pagination": {
  "page": 1,
  "limit": 20,
  "total": 12,
  "total_pages": 1
}
}
```

6.2 Generate Report

POST /api/v1/reports

Request Body

Field	Type	Required	Description
title	string	Yes	Report title
type	string	Yes	daily, weekly, monthly, custom
date_from	string	Yes	ISO 8601 start date
date_to	string	Yes	ISO 8601 end date
camera_ids	string []	No	Restrict to specific cameras
format	string	No	pdf (default) or csv

```
{
  "title": "Monthly Incident Overview - July",
```

```
"type": "monthly",
"date_from": "2026-07-01T00:00:00Z",
"date_to": "2026-07-31T23:59:59Z",
"format": "pdf"
}
```

Response 202 Accepted

```
{
  "id": "rep_090",
  "status": "processing",
  "requested_at": "2026-07-29T10:20:00Z"
}
```

Errors: 400 Bad Request

6.3 Get Report

GET /api/v1/reports/{id}

Path Parameters: id (string, required)

Response 200 OK

```
{
  "id": "rep_090",
  "title": "Monthly Incident Overview - July",
  "type": "monthly",
  "status": "completed",
  "generated_at": "2026-07-29T10:22:00Z",
  "file_url": "https://cdn.yourdomain.com/reports/rep_090.pdf"
}
```

Errors: 404 Not Found

7. AI Models

7.1 List Models

GET /api/v1/models

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number
limit	integer	20	Items per page
type	string	-	object_detection, facial_recognition, motion_detection, etc.
status	string	-	active, inactive, training

Response 200 OK

```
{
  "data": [
    {
      "id": "model_007",
      "name": "Perimeter Intrusion Detector",
      "type": "object_detection",
      "version": "2.3.1",
      "status": "active",
      "assigned_cameras": ["cam_012", "cam_013"]
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 6,
    "total_pages": 1
  }
}
```

7.2 Register Model

POST /api/v1/models

Request Body

Field	Type	Required	Description
name	string	Yes	Model display name
type	string	Yes	Model category
version	string	Yes	Version identifier
confidence_threshold	number	No	Minimum confidence (0.0–1.0), default 0.7
assigned_cameras	string[]	No	Cameras to activate the model on

```
{  
  "name": "Loitering Detector",  
  "type": "behavior_analysis",  
  "version": "1.0.0",  
  "confidence_threshold": 0.8,  
  "assigned_cameras": ["cam_012"]  
}
```

Response 201 Created

```
{  
  "id": "model_008",  
  "name": "Loitering Detector",  
  "type": "behavior_analysis",  
  "version": "1.0.0",  
  "status": "training",  
  "created_at": "2026-07-29T10:25:00Z"  
}
```

Errors: 400 Bad Request

7.3 Update Model

PATCH /api/v1/models/{id}

Path Parameters: id (string, required)

Request Body (partial update)

Field	Type	Description
status	string	active, inactive, training
confidence_threshold	number	Minimum confidence (0.0-1.0)
assigned_cameras	string[]	Cameras to activate the model on

```
{
  "status": "active",
  "confidence_threshold": 0.75
}
```

Response **200 OK** - returns the updated model object.

Errors: 400 Bad Request

404 Not Found

8. Notifications

8.1 List Notifications

GET /api/v1/notifications

Query Parameters

Param	Type	Default	Description
page	integer	1	Page number

limit	integer	20	Items per page
read	boolean	-	Filter by read/unread status

Response **200 OK**

```
{
  "data": [
    {
      "id": "not_301",
      "title": "New critical alert",
      "message": "Critical alert triggered on Front Entrance camera.",
      "read": false,
      "created_at": "2026-07-29T09:58:05Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 3,
    "total_pages": 1
  }
}
```

8.2 Create Notification

POST /api/v1/notifications

Request Body

Field	Type	Required	Description
title	string	Yes	Notification title
message	string	Yes	Notification body
recipients	string[]	Yes	User IDs to notify
channel	string	No	in_app, email, sms (default in_app)

```
{
  "title": "Scheduled Maintenance",
  "message": "Camera cam_013 will be offline for maintenance at 2 PM.",
  "recipients": ["usr_001", "usr_045"],
  "channel": "email"
}
```

Response 201 Created

```
{
  "id": "not_302",
  "title": "Scheduled Maintenance",
  "status": "sent",
  "created_at": "2026-07-29T10:30:00Z"
}
```

Errors: 400 Bad Request

8.3 Mark Notification as Read

PATCH /api/v1/notifications/{id}/read

Path Parameters: **id** (string, required)

Response 200 OK

```
{
  "id": "not_301",
  "read": true,
  "read_at": "2026-07-29T10:31:00Z"
}
```

Errors: 404 Not Found

9. Endpoints

Resource	Endpoints
Authentication	POST /api/v1/auth/login, POST /api/v1/auth/refresh, POST /api/v1/auth/logout
Users	GET /api/v1/users, POST /api/v1/users, GET /api/v1/users/{id}, PUT /api/v1/users/{id}, DELETE /api/v1/users/{id}
Cameras	GET /api/v1/cameras, POST /api/v1/cameras, GET /api/v1/cameras/{id}, PATCH /api/v1/cameras/{id}, DELETE /api/v1/cameras/{id}
Alerts	GET /api/v1/alerts, GET /api/v1/alerts/{id}, PATCH /api/v1/alerts/{id}
Incidents	GET /api/v1/incidents, POST /api/v1/incidents, GET /api/v1/incidents/{id}, PATCH /api/v1/incidents/{id}
Reports	GET /api/v1/reports, POST /api/v1/reports, GET /api/v1/reports/{id}
AI Models	GET /api/v1/models, POST /api/v1/models, PATCH /api/v1/models/{id}
Notifications	GET /api/v1/notifications, POST /api/v1/notifications, PATCH /api/v1/notifications/{id}/read

Common Response Formats

Pagination Object

All list endpoints return a **pagination** object:

```
{
  "page": 1,
  "limit": 20,
  "total": 42,
  "total_pages": 3
}
```

Error Object

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "The 'email' field is required.",
    "details": {
      "field": "email"
    }
  }
}
```

Error Codes

HTTP Status	Code	Description
400	VALIDATION_ERROR	Request body failed validation
401	UNAUTHORIZED	Missing or invalid authentication token
403	FORBIDDEN	Authenticated but lacks permission for this action
404	NOT_FOUND	Requested resource does not exist
409	CONFLICT	Request conflicts with current resource state

422	UNPROCESSABLE_ENTITY	Well-formed request but semantically invalid
429	RATE_LIMITED	Too many requests; retry after backoff
500	INTERNAL_ERROR	Unexpected server error

Rate Limiting

All endpoints are subject to rate limiting. Current limits are returned in response headers:

X-RateLimit-Limit: 100

X-RateLimit-Remaining: 87

X-RateLimit-Reset: 1753786800

Exceeding the limit returns **429 Too Many Requests** with a **Retry-After** header (seconds).

Versioning

This API is versioned via the URL path (**/api/v1/...**). Breaking changes will be introduced under a new version prefix (e.g. **/api/v2/...**); non-breaking additive changes may be released within **v1**.